CPS1102 Fall 2023 SPSS Notes & Examples

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Contents

1 Preface

This document is meant to serve the following purposes for CPS1102:

1. To provide referential and supplemental notes to the material covered in lecture
2. To provide the steps in **SPSS** required to run the various statistical methods covered in this class
3. To provide some elaboration regarding the output of results from SPSS
4. Hopefully provide some insight into data cleaning and exploration in SPSS

Throughout this document, I make use of *real* datasets from published clinical psychology papers. These datasets are the same ones that are used during class demonstrations. I provide an overview of the actual datasets and relevant variables, where needed, along with citations to the papers that the data come from.

Many of the explanations for the statistical methods outlined in this document are somewhat informal in nature, and are not meant to be fully rigorous. Please refer to the appropriate class slides for more information.

Any inconsistencies or errors with respect to grammar or content are my own doing (but I did my best to proofread).

Please contact me if you have any questions or concerns regarding any of the material in this document!

2 Data Preparation/Cleaning

2.1 Background

For this section of the document, we will make use of the `SymptomData.csv` dataset.

2.2 SPSS

As mentioned in lecture, it is useful to have a good handle on the following things with respect to data preparation/cleaning:

- Subsetting datasets (either subsetting participants or variables)
- Recoding variables
- Creating new variables out of existing variables
- Calculating z-scores
- General descriptive statistics (numerical and graphical)
- Evaluating and handling missing data

2.2.1 Subsetting Datasets

In some cases, you might either want to **subset** a dataset depending on some inclusion/exclusion criteria, or you might want to **merge** two datasets if you happened to collect more participants or more variables. Subsetting a datset can be done on a variable-basis (i.e., removing certain variables that aren't of interest) or on a participant-basis (i.e., removing certain participants that meet certain criteria or scores on variables).

The following example will demonstrate how to subset the `SymptomData.csv` dataset on a participant-basis. In particular, we will subset (i.e., filter) participants who have a score greater than 10 on the PHQ-9 - that is, we only want these participants in our dataset. The steps are as follows:

1. Data -> Select Cases
2. Click `If condition is satisfied`, and then `If` below that
3. In the new box that pops up, drag the `phq_total` variable into the top right box, and further type in '>10' using the buttons provided; click `Continue`
4. If you would like to maintain the same dataset window open in SPSS, keep the `Filter out unselected cases` option checked. Otherwise, you can create a new dataset with these separate participants by clicking `Copy selected cases to a new dataset` and giving the dataset a name

2.2.2 Creating New Variables and Recoding

Typical cases of variable creation and recoding in psychology include:

- Converting a continuous variable (e.g., total scores) into a binary (e.g., those who meet a cut-off vs those who don't)
- Taking the sum or mean of items in a questionnaire and creating an associated total score
- Creating z-scores

2.2.2.1 Continuous -> Binary Here, we are going to create a new *binary* variable in SPSS. In particular we are going to create a variable representing 'probable depression diagnosis' using scores on the PHQ-9. You will see in the dataset that a variable like this seems to be already in there (given the name `dep_dx`), but we will create our own version of this variable for verification and for the purpose of example.

According to scoring instructions and existing research, scores of 10 or greater seem to indicate a probable depression diagnosis (e.g., https://jamanetwork.com/journals/jama/article-abstract/2766865). Below, we create a new variable called `dep_dx_again` which takes on a value of 1 if a participant has a score of 10 or greater on `phq_total`, and 0 otherwise.

1. Transform -> Recode into Different Variables
2. Drag `phq_total` into the right box and click `Old and New Values`

3. Click `Range, value through HIGHEST` and enter 10 (i.e., it asks for what score value and *greater* you want to recode; in our case, we want to recode values of 10 or greater to a single value)
4. Under `New Value` type in 1 (meaning those with phq_total values greater than or equal to 10 receive a new response of 1 in the new variable we are creating) -> Click `Add`
5. Click `Range, LOWEST through value` and enter 9 (i.e., it asks for what score value and *lower* you want to recode; in our case, we want to recode values of 9 or less to a single value)
6. Under `New Value` type in 0 (meaning those with phq_total values less than or equal to 9 receive a new response of 0 in the new variable we are creating) -> Click `Add` and `Continue`
7. Click the variables in the middle and then type in the new variable name `dep_dx_again` -> Click `Change` -> Click `OK`

Now, the new variable can be found in the dataset.

A similar procedure can be done if you would like to recode *individual* values of a variable (e.g., changing every instance of 2 to 1, every instance of 3 to 2, etc.). Instead of using the `Range` options under the `Old and New Values` section, you would recode value by value in the `Old and New Values` box that pops up. Specifically, you would make use of the `Value` section under `Old Value`, instead of the `Range` options.

2.2.2.2 Total Scores Here, we are going to create a new total score variable in SPSS. In particular, we are going to create a total score using PHQ-9 items. Again, there is already a variable in the dataset that is the sum score of all of the PHQ-9 items (`phq_total`), but in the following example we will create a total score that only involves two PHQ-9 items (the first and second). Before this is done, however, we have to make sure that the item variables themselves are of the appropriate scale type. At the moment, they are coded as 'Nominal' but they need to be changed to 'Scale' in order for addition to actually be done on them. Once that is done, we can compute the new total score:

1. Transform -> Compute Variable
2. Under `Target Variable`, type in the name you want to give to the new variable (in this case, `phq_total_again`)
3. Insert `phq1` into the top right box, click `+`, then insert `phq2` right beside it
4. Click `OK`

The new variable can now be found in the dataset.

2.2.2.3 Creating Z-Scores To create z-scores for an existing variable in SPSS:

1. Analyze -> Descriptive Statistics -> Descriptives
2. Drag the variable(s) you want to create z-scores for to the right box and also click `Save standardized values as variables`
3. Click `OK` and if you go back to the dataset you should see a new variable corresponding to z-scores for your variables of interest

2.2.3 Variable Descriptives and Plots

When first looking at a dataset, it is helpful to look at various descriptive statistics for the purposes of finding strange values, possible errors in data input, and to generally get a sense of what the different variables look like. As mentioned in other lectures, another important thing to keep in mind when looking at a dataset for the first time is to see whether the variable *types* are what they should be.

When the `SymptomData.csv` dataset is loaded into SPSS, you will see that, for example, the `gad_total` variable is coded as 'Nominal' rather than 'Scale'. To change this variable to Scale (i.e., a continuous variable), simply click the variable type for the corresponding variable and switch it to Scale.

Now, let's look at an overview of descriptive statistics for some variables in the dataset. As alluded to above, there are a couple of things that might be worth paying attention to. First, in the case of quantitative variables, you want to see whether the minimum and maximum values match with expectations, as this is one way to determine whether there are coding errors or strange values. For example, if it is expected that phq_total should only take values from 0 to 27, it should be checked that there aren't any values below or above this range by looking at the minimum and maximum values. In the case of categorical variables, it helps to look at every possible category that the variable can take on. Second, with quantitative variables, skew and kurtosis values can also give you some insight into how much the variables may be close to a normal distribution.

To look at descriptive statistics for continuous variables in SPSS:

1. Analyze -> Descriptive Statistics -> Descriptives
2. Drag the variables you want to describe to the right box
3. Click `Options` and select Kurtosis and Skewness
4. Click `Continue` and then `OK`

A similar procedure can be done for categorical variables: instead of clicking `Descriptives`, click `Frequencies`; select the statistics you would like to calculate under the `Options` button; click `Continue` then `OK`.

Histograms can be used to check distributions of variables and see whether there are potential outliers:

1. Graphs -> Chart Builder
2. Click Histogram and drag it into the Chart Preview
3. Drag the variable you want to plot onto the x-axis; click `Display normal curve`; click `OK`

An alternate way to do the above, which provides additional information regarding outliers and normality, is to do the following:

1. Analyze -> Descriptive Statistics -> Explore
2. Drag the variable of interest into the `Dependent List` box
3. Click `Statistics`, select `Outliers`, click `Continue`
4. Click `Plots`, select `Histogram` and `Normality plots with tests`, click `Continue`
5. Click `OK`

2.2.3.1 Outliers The above procedures are fine with respect to visual inspections of outliers. Traditional recommendations for outlier detection include declaring a participant an outlier on a variable if their z-score on that given variable exceeds $\pm$ 3. This procedure is not generally recommended as it is not a robust method. An alternate method is one that uses medians and median absolute deviations, as demonstrated in this article: https://www.sciencedirect.com/science/article/pii/S0022103113000668

Using the phq_total variable as an example, to determine outliers using this robust method in SPSS:

1. For the variable of interest, go to Analyze -> Descriptive Statistics -> Frequencies; put the variable in; uncheck `Display frequency tables`; click `Statistics` and select 'Median'; click `OK`
2. Take note of this median value (5)
3. Transform -> Compute Variable; in Target Variable type in 'out1_phq_total' (can put whatever you want here); under Numeric Expression type in `abs(phq_total - 5)` where 5 is the value of the median from step 2; Click `OK`
4. Analyze -> Descriptive Statistics -> Frequency; put the 'out1_phq_total' variable in; click `Statistics` and select 'Median'; click `OK`
5. Take note of *this* median value (3)
6. Multiply this median value by 1.4826 (3 times 1.4826 = 4.4478)
7. Transform -> Compute Variable; in Target Variable type in 'outlier_phq_total'; under Numeric Expression type in `(out1_phq_total - 5)/(4.4478)`; Click `OK`

Now, you have a variable ('outlier_phq_total') that has scores for each person, where these scores can then be used to determine whether a person meets a cutoff for being an outlier. In the paper cited above, they recommend a cutoff value of 2.5. To remove participants that are above this cutoff:

1. Data -> Select Cases

2. Click 'If' under `If condition is satisfied`; drag `outlier_phq_total` to the right, type in `> 2.5`; click `Continue`
3. Click `Copy selected cases to a new dataset`, name the new dataset, then click `OK`

A new dataset should be created that contains those who were not considered outliers using the cutoff value of 2.5. When reporting results, it is important to note *how many* participants were excluded due to being outliers, and what the resulting new dataset sample size is.

2.2.4 Missing Data

To examine missing data (e.g., descriptives, MCAR tests, t-tests) and do *imputation* at the same time (i.e., EM imputation):

1. Analyze -> Missing Value Analysis
2. Drag the variables of interest into the appropriate boxes to the right
3. Click `Descriptives`, select the t-tests option, select 'Include probabilities in table', and click `Continue`
4. Under Estimation click the EM checkbox
5. Then click the EM button, and select `Save completed data` so the imputed dataset is made and is separated from the original data
6. Click `OK`

When this is done, a new dataset should pop up along with the output of the missing data analyses. In the missing data analyses, you will find Little's MCAR test (i.e., if it is significant, this may mean that the data are not MCAR), along with the missing data t-tests. The purpose of the missing data t-tests is to see whether missingness on one variable is systematically (i.e., significantly) related to values on another variable, which is a characteristic of MAR. In the t-test table, the rows represent the variable one is testing missingness patterns for, and the columns represent the other variables in the dataset. That is, the t-test table shows whether missingness on a row variable is associated with values on a column variable.

3 Correlations

3.1 Background

There are several types of correlations that we covered in class that can be run in SPSS. ALl of these correlations commonly aim to quantify the degree of an association between two variables, but differ in their underlying calculations, assumptions, and even the research question they can answer.

Arguably, the most popular type of correlation that is used in clinical psychology is **Pearson's correlation coefficient**. Pearson's correlation quantifies the degree of a *linear* association between two variables. In other words, it tells you how likely it would be to have an above average score on one variable (relative to those in the sample) if you also happen to be above average on the other variable (at least in the case of a *positive* correlation). A *negative* correlation would represent the likelihood of having a below average score on one variable while also having an above average score on the other variable. For example, a negative correlation between depression and relationship quality would mean that those who have above average depression are also likely to have below average relationship quality. The main parameter of interest with correlations is the actual correlation coefficient, represented by r and ranges from -1 (perfectly negative correlation) to +1 (perfectly positive correlation).

As discussed in class, Pearson's correlation is appropriate to use when there is a hypothesized linear association, when the two variables in question are continuous, and when there is *bivariate normality* in the joint distribution of the two variables together. These characteristics can and will be examined using SPSS.

Alternate correlation coefficients, when the above characteristics are not met, are also possible in SPSS. Some of these are more appropriate for different types of variables (e.g., Kendall's correlation with ordinal variables), some are able to detect nonlinear but increasing associations (e.g., Spearman's), and some are robust to outliers and nonnormal distributions.

Even other types of correlations are demonstrated below, including the semipartial and partial correlations (correlation between two variables while controlling for the effect of a third variable).

Below, we will use the `SymptomData.csv` dataset to estimate correlations.

3.2 SPSS

3.2.1 Pearson's Correlation

Let's take a look at the correlation between depression (variable: phq_total) and relationship quality (variable: rq_total).

1. Analyze -> Correlate -> Bivariate
2. Select the two variables that you want to correlate
3. Click `Confidence interval` -> check off the first box -> click `continue`
4. For a cleaner output, on the bottom of the dialogue box click `Show only the lower triangle` and deselect `Show diagonal`
5. Click `OK`

3.2.1.1 Checking Assumptions First, to check the assumption of **linearity**, it is helpful to view the two variables in a scatterplot, along with a LOESS curve.

To plot a scatterplot for the correlation between two variables:

1. Graphs -> Chart Builder 2
2. Go to Scatter/Dot, drag the first graph into the chart preview
3. Drag the two variables onto the x and y axes
4. Near the bottom right of the dialogue box, click `Total` under `Linear Fit Lines`
5. Click `OK`

This is almost always a good idea to do *before* you estimate a correlation in order to see whether there are any nonlinear patterns or extreme outliers.

Another interesting graph you can do is called a LOESS plot, which applies a more flexible curve to the scatterplot depending on what the data look like, rather than a strict straight line. Seeing the output of a LOESS plot can help determine whether there are substantial nonlinear relations between two variables. To create a LOESS plot in SPSS:

1. Graphs -> Chart Builder
2. Go to Scatter/Dot, drag the first graph into the chart preview
3. Drag the two variables onto the x and y axes -> Click `OK`
4. Double click the resulting graph -> At the top right click `Add Fit Line at Total` -> Select `LOESS`

Second, to check the assumption of **normality** and **lack of outliers**, the distributions of each variable can be visualized and tested against a normal distribution.

1. Analyze -> Descriptive Statistics -> Explore
2. Put the variables in the 'Dependent List' box
3. Under Plots, select 'Histogram' and 'Normality plots with tests'

3.2.2 Spearman's Rank Correlation

1. Analyze -> Correlate -> Bivariate
2. Select the two variables that you want to correlate
3. Select `Spearman` under `Correlation Coefficients`
4. Click `Confidence interval` -> check off the first box -> click `continue`
5. For a cleaner output, on the bottom of the dialogue box click `Show only the lower triangle` and deselect `Show diagonal`
6. Click `OK`

3.2.3 Kendall's Tau Correlation

1. Analyze -> Correlate -> Bivariate
2. Select the two variables that you want to correlate
3. Select `Kendall's tau-b` under `Correlation Coefficients`
4. Click `Confidence interval` -> check off the first box -> click `continue`
5. For a cleaner output, on the bottom of the dialogue box click `Show only the lower triangle` and deselect `Show diagonal`
6. Click `OK`

3.2.4 Bootstrapping Correlations

To derive bootstrapped confidence intervals for any of the above correlation coefficients:

1. Analyze -> Correlate -> Bivariate
2. Select the two variables you want to correlate along with the type of correlation
3. Click `Bootstrap`, select `Perform bootstrapping`, set the number of bootstrapped samples you want
4. Select `Set seed` and put in a random number to ensure reproducibility
5. Under CIs, select BCa
6. Click `Continue` and then `OK`

Note that bootstrapping can also be done with semipartial and partial correlations.

3.2.5 Semipartial and Partial Correlations

1. Analyze -> Regression -> Linear
2. Put in the main dependent variable and independent variable, along with the other independent variables you'd like to control for

3. Click Statistics and select Part and partial correlations
4. Click Continue and then OK

In the output, you should see the semipartial (labeled as part) and partial correlations.

3.2.6 Correlation Matrices

If you would like to estimate pairwise correlations between more than two variables, this information can be presented in a correlation matrix.

1. Analyze -> Correlate -> Bivariate
2. Select all of the variables that you want to correlate
3. Select your preferred correlation type under Correlation Coefficients
4. Click Confidence interval -> check off the first box -> click continue
5. For a cleaner output, on the bottom of the dialogue box click Show only the lower triangle and deselect Show diagonal
6. Click OK

3.3 Sample Size Planning

Power analyses for Pearson's correlations can be conducted using the G*Power software, see here (https://www.psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower) and here as well (https://link.springer.com/article/10.3758/BRM.41.4.1149). This website provides a manual for conducting power analyses for correlations in this software (among many other statistical tests). G*Power can also do power analyses for *differences* in correlation values.

Another useful website that can be used for correlation power analyses can be found here: https://sample-size.net/correlation-sample-size/. This website lets you input a given alpha value, your Type II error rate (which is 1 - statistical power; e.g., for 80% statistical power, the Type II error rate would be 1 - .80 = .20), and the size of the correlation you would like to detect.

For Spearman's and Kendall's correlations, to my knowledge there is no readily available software for power analyses, but this paper may be useful as it gives benchmarks for sample size planning given different values of Spearman's and Kendall's correlations: https://www.hilarispublisher.com/open-access/sample-size-charts-of-spearman-and-kendall-coefficients.pdf.

4 Regression (Simple and Multiple)

4.1 Background

In the following examples, we will use the same dataset that was used for the correlations notes - the `SymptomData.csv` dataset.

In lecture, we covered both simple linear regression (a single independent variable) and multiple linear regression (more than one independent variable). The example that we used for **simple linear regression** involved depression as the *independent variable* and relationship quality as the *dependent variable*. The example that we used for **multiple linear regression** involved depression, anxiety, and social phobia as the independent variables with relationship quality as the dependent variable.

4.2 SPSS

4.2.1 Simple Linear Regression

Here, we will run a simple linear regression model where we would like to predict relationship quality from depression scores only.

1. Analyze -> Regression -> Linear
2. Enter the appropriate independent variables (in this case, IV is phq_total and DV is rq_total)
3. Click `Statistics` and select `Confidence intervals`; click `Continue`
4. Click `OK`

In the output, there are a few tables of interest. First, the `Model Summary` table provides R^2. Second, the `ANOVA` table provides the model F-test for testing whether R^2 is significantly different from 0. Third, the `Coefficients` table provides the betas (both unstandardized and standardized) along with respective standard errors and confidence intervals.

4.2.1.1 Checking Assumptions There are a couple different sets of steps that need to be run in order to check the various assumptions involved in a regression model.

1. Complete the first two steps for Simple Linear Regression above
2. Select `Statistics` and select `Durbin Watson` under `Residuals`; click `Continue`
3. Click `Plots` and select `Histogram` and `Normal probability plot` under Standardized REsidual Plots; drag `*ZRESID` to Y and `*ZPRED` to X; click `Continue`
4. Click `Save` and select Unstandardized (under Predicted Values), each of the distances, Standardized, Studentized, and Studentized deleted residuals (under Residuals), and Standardized DfBetas and DfFits (under Influence Statistics); click `Continue`
5. Click `OK`

First, let's see how we assess the **assumption of linearity**. In the output, there should be a scatterplot that has predicted values on the x-axis and residuals on the y-axis. The dots on the plot should not be moving upwards or downwards. A way to double-check this is to create a LOESS curve on the plot (double-click, select Add Fit Line at Total button in top right, select LOESS). The LOESS curve should be relatively straight. Yet another way to check linearity, at least in the case of *simple* linear regression, is to create a scatterplot of the two variables of interest and make a LOESS curve on this plot (see the Correlations section of the document for how to do this).

Second, let's see how we assess the **assumption of constant variance**. We can take a look at the same scatterplot that we looked at for the linearity assumption. In this case, however, we want the vertical 'spread' of the residuals to be relatively similar going from left to right. To assess this assumption with a statistical test (while keeping in mind limitations of such an approach), a statistical test that can be used is the Breusch-Pagan test. To conduct a Breusch-Pagan test:

1. Analyze -> General Linear Model -> Univariate

2. Put the variables of interest in the appropriate boxes as in the steps above (i.e., independent variables would go in `Covariate`)
3. Click `Options` and select Breusch-Pagan test; click `Continue`
4. Click `OK`

Similar to the Shapiro-Wilk's test, the idea is that if this test is statistically significant, then it might be the case that the constant variance assumption is violated.

Third, let's see how we assess the **assumption of normality**. Going back to the original output from the five steps above, there will be two relevant plots: one with the histogram of residuals and a P-P plot. The histogram of residuals should look relatively normally distributed. In the P-P plot, the dots should be roughly along the straight diagonal line. Because the residual values were saved, a variation of this previous visualization can be done. Go to Analyze -> Descriptive Statistics -> Explore, then drag the Standardized Residual variable into the `Dependent List`, click `Plots` and select Normality plot with tests, then click `Continue` and `OK`. Here, a Shapiro-Wilk's test will be run on the distribution of residuals.

Fourth, let's see how we assess the **assumption of independence**. While this is mostly a study design issue, it can also be formally checked with the Durbin-Watson test that was specified above. In the output of SPSS, the value of the Durbin-Watson test should be *around* 2 and not greater than 3 or less than 1.

Finally, let's see how we assess for **outliers**, which is not necessarily an assumption of a regression model but the presence of such can negatively affect results. To check for outliers, you can take a look at descriptives of the model statistics that you saved above (i.e., distances, residuals, influence statistics). Similar to what was done with the Standardized Residuals, to look at outliers:

1. Analyze -> Descriptive Statistics -> Explore
2. Drag Studentized Deleted Residual, Mahalanobis Distance, Cook's Distance, Leverage Value, DfFit, and DfBeta into `Dependent List`
3. Click `OK`

In the output you will see descriptive statistics for each of these values, the top 5 lowest and highest values on each of these variables, and boxplots that can indicate outliers. As a reminder: high leverage values and Mahalanobis distance has to do with outliers in the IVs, studentized residuals have to do with outliers in terms of residuals, and influence (Cook's D, DfFits, DfBeta) has to do with outliers that affect parameter estimates. Remember that you can also conduct the outlier methods that were outlined in the DataPrep section of the document with these values here. For more information on recommended cut-off values for these indices, please see the lecture slides.

With Mahalanobis Distance in particular, the values that are calculated in SPSS can be compared to values in a statistical table to determine which cases appear to be 'significant' outliers:

1. Go here: https://www.itl.nist.gov/div898/handbook/eda/section3/eda3674.htm and scroll to the first table
2. V corresponds to the number of IVs you are considering and should look at the .99 column
3. If a participant's Mahalanobis Distance value is above this value then they might be considered an outlier

Another informative plot that can be created is one that has Leverage values on the x-axis and Standardized Residuals on the y-axis. If a particular participant has high values on both of these variables (i.e., their data point is in the top right corner of this graph), then there is evidence that this participant may be an outlier. To create this chart:

1. Go to Chart Builder, create a scatterplot, and drag Leverage to the x-axis and Standardized Residuals to the y-axis; click `OK`
2. To determine which participant is which data point, you can double click on the chart, click `Data Label Mode` at the top right, and click on the data points to determine the participant.

4.2.2 Multiple Linear Regression

Here, we will run a multiple regression model where we would like to predict relationship quality from depression, anxiety, and social phobia scores. The procedure to estimate a multiple regression model is very similar to that of simple linear regression.

The steps are basically the same, with the addition of adding more variables to the independent variables box.

4.2.2.1 Checking Assumptions All of the assumptions that apply to simple linear regression (described above) also apply here to multiple linear regression. The additional condition that should be checked that is unique to multiple regression is *collinearity*. To get collinearity diagnostics:

1. Go through the specification of a multiple regression model
2. Select `Statistics`, and check the `Collinearity diagnostics` box; click `Continue`
3. Click `OK`

4.2.3 Hierarchical Regression

Hierarchical regression is a methodology where you are interested in comparing the predictive performance of a larger model to the predictive performance of a smaller model. In some psychological research, this is often called 'incremental validity', and basically tries to answer whether an additional independent variable(s) explains a significant amount of more variance in an outcome *above and beyond* other independent variables already accounted for. For example, we can see whether the addition of anxiety and social phobia scores provide incremental validity for predicting relationship quality, above and beyond depression scores. Note, hierarchical regression is not to be confused with *hierarchical linear models*, which is another name for multilevel models.

To run a hierarchical regression in SPSS:

1. Analyze -> Regression -> Linear
2. Put the phq_total and rq_total variables in the appropraite spots
3. Click `Next` to create a 'second block', add the gad_total and sp_total variables here
4. Under `Statistics`, select `R-squared change`; click `Continue` and click `OK`

Under the `Model Summary` table in the output, the p-value assogiaged with the R^2 change can be found.

4.2.4 Robust Versions of Regression

There are a couple of different methods that can be employed to account for particular violated assumptions. If it is the case that the assumption of **constant variance** is severaly violated, there are two methods that can be done in SPSS. The **first** method is to create bootstrapped confidence intervals for the betas. To create bootstrapped confidence intervals:

1. Go through the specification of a regression model as usual
2. Select `Bootstrap`, click `Perform bootstrapping`, set the number of samples to a high number (1000 is usually fine but can do up to 5000)
3. Click `Set seed` and enter a number so that the randomization involved in creating the bootstrapped samples is reproducible if you were to run this analysis again
4. You can select either `Percentile` or `BCa` under confidence intervals; click `Continue`
5. Click `OK`

The **second** method is to use so-called 'robust' standard errors which correct for the constant variance violation and produce more accurate confidence intervals and p-values. To do this in SPSS:

1. Analyze -> General Linear Model -> Univariate
2. Put the dependent variable of interest unde `Dependent Variable` and the independent variables of interest under `Covariate`
3. Click `Options` and select `Parameter estimates with robust standard errors`; click `Continue`

4. Click `OK`

If it is the case that the assumption of *normally distributed residuals* is severely violated, or if there are residual outliers, one method that can be used is called Quantile Regression. This method is similar to a typical regression model, but takes into account medians rather than means, as medians are more robust to nonnormality and outliers. To run a quantile regression in SPSS:

1. Analyze -> Regression -> Quantile
2. Put the dependent variable of interest under `Target Variable` and the independent variables of interest under `Covariate(s)`
3. Click `OK`

In the output, you will get the betas for your model, along with standard errors and confidence intervals for the betas.

4.3 Sample Size Planning

Conducting a power analysis for regression models can be tricky because one can conduct their power analysis based on detecting a R^2 value of interest (i.e., a certain amount of variance explained) or based on detecting a particular regression coefficient in the model. Both choices, however, can be done in software.

Two resources for conducting a power analysis for regression model `G*Power` are the following:

- https://web.pdx.edu/~newsomj/mvclass/ho_sample%20size.pdf
- https://link.springer.com/article/10.3758/BRM.41.4.1149

THere are also online applications, built from R, that may be easier to use without having to write out R code:

- https://designingexperiments.shinyapps.io/BUCSS_ss_power_reg1/ (an online app for the power of a single coefficient)
- https://designingexperiments.shinyapps.io/BUCSS_ss_power_reg_all/ (an online app for the power of R^2)

5 T-Test and One-Way ANOVA

5.1 Background

In the following examples, we will make use of two different datasets for t-tests and the one-way ANOVA, respectively. The dataset that will be used for the t-tests is the familiar `SymptomData.csv` dataset that has been used for the previous statistical methods. The dataset that will be used for the one-way ANOVA is a new dataset - `EmotionRegulation.csv` - which was described in the slides for the t-test/ANOVA lecture.

All of these methods are commonly concerned with testing the null hypothesis of equality of means either between two or more separate groups (independent-samples t-test, ANOVA), or with respect to a given value (one-sample t-test).

5.2 SPSS

5.2.1 Independent-Samples T-Test

Here, we will conduct an independent-samples t-test to determine whether the mean of relationship quality scores in those with MDD is significantly different than the mean of relationship quality scores in those without MDD.

5.2.1.1 Preliminary Assumption Checking In this particular dataset, the grouping variable of MDD diagnosis needs to be slightly recoded so SPSS knows how to handle the categorical variable. This only needs to be done if there isn't already explicit coding in the dataset that associates each category with a particular number. First, we will recode the `dp_dx` variable by:

1. Transform -> Recode into Same Variables
2. Drag the `dp_dx` variable into the Variables list
3. Click `Old and New Values`; type 'No MDD' under 'Old Value/Value:' and then type 0 under New Value; click Add; repeat for 'Yes MDD' except type 1 under New Value instead
4. Click `OK`
5. Go under Variable View and change the *variable type* (the second column) to numeric.

For assumptions, one should check whether the assumption of **normality** (i.e., normal residuals) is not severely violated. Examining the residuals of this test is equivalent to looking at the histograms of the raw outcome variable *within* each group separately. For example, in this case, you would look at the histogram/descriptives of `rq_total` for those in the 'Yes MDD' group, and separately `rq_total` for those in the 'No MDD' group. To get this information (and more):

1. Analyze -> Explore
2. Put the outcome variable in the `Dependent List` box, and the grouping variable under `Factor List`
3. Under `Plots`, click 'Histogram' and 'Normality plots with tests'; click `Continue`
4. Click `OK`

In the output, you should get group-wise descriptive statistics for the outcome variable, tests of normality, and plots of normality.

With respect to the constant variance assumption, it's recommended in the literature to simply not test for this assumption and by default look at the results for Welch's t-test, which does not assume equal variances between groups.

5.2.1.2 Actual Test Now, to actually conduct the independent-samples t-test:

1. Analyze -> Compare Means -> Independent-Samples T-Test
2. Put the outcome of interest in the `Test Variable(s)` box
3. Put the grouping variable in the `Grouping Variable` box
4. Check off `Estimate effect sizes`; Click `OK`

The output of the test should include three different tables. The first table provides descriptive statistics for the outcome variable of interest stratified by group. The second table provides the actual significance test. Importantly, the second row that says `Equal variances not assumed` is where the Welch's t-test is, and this should be focused on. The third table provides the associated effect sizes with the test.

To plot the results, go to Graphs -> Chart Builder, select a bar plot, drag the variables to their appropriate spots (i.e., grouping variable should be on x-axis), and check off `Display error bars` on the side.

5.2.2 One-Way ANOVA

Here, we will make use of the `EmotionRegulation.csv` dataset. We will conduct a one-way ANOVA to determine whether the mean of negative affect scores are significantly different between the healthy control group, the MDD group, and the BPD group.

5.2.2.1 Preliminary Assumption Checking Similar to what was described above under the independent-samples t-test, the grouping variable needs to be recoded in this case so that SPSS can associate a particular number to each category. Here, we will code 'ctl' as 0, 'mdd' as 1, and 'bpd' as 2.

The assumption of normality can be examiend using the same procedure as in the independent-samples t-test. That is, one can go to Analyze -> Explore, put in the appropriate variables (i.e., outcome in Dependent List, grouping variable in Factor List), and under Plots select Histogram and Normality plots. With respect to the constant variance assumption, similar to what is recommended for independent-samples t-test, you can look at the Welch's one-way ANOVA which can be estimated using the steps below.

5.2.2.2 Actual Test To run the one-way ANOVA:

1. Analyze -> Compare Means -> One-Way ANOVA
2. Put the appropriate variables in their respective boxes (i.e., the outcome goes under 'Dependent List' and the grouping variables goes under 'Factor')
3. Check off `Estimate effect sizes for overall tests`
4. Under `Post Hoc` you can select the post-hoc tests that you would like. Typical tests that are used are the LSD and Bonferroni (equal variances assumed), and Games-Howell and the Dunnett's tests (equal variances not assumed); Click `Continue`
5. Under `Options`, select Descriptives, Homogeneity of variance test, Welch test, and Means plot; Click `Continue`
6. Click `OK`

In the output, the first table provides the group-wise descriptive statistics for the outcome variable. The second table provides Levene's test for constant variance, but as mentioned above and in lecture, this can be ignored if you would like to just use the robust Welch's one-way ANOVA. The third table provides the overall F-test for the ANOVA. The frouth table provides global effect sizes associated with the one-way ANOVA. The fifth table provides the robust Welch's one-way ANOVA, which is basically the robust version of the F-test found in the third table. The last table provides the post hoc tests between the actual groups. A plot is also provided that shows the means for the outcome variable stratified by group. You can also plot the results using the same procedure as in the independent-samples t-test.

5.2.3 One-Sample T-Test

Here, using the `EmotionRegulation.csv` dataset, we will conduct a one-sample t-test to determine whether the mean of negative affect scores is significantly different from a value of 1.

5.2.3.1 Preliminary Assumption Checking One should check whether the assumption of normality (i.e., normal residuals) is not severely violated. In the case of a one-sample t-test, examining residuals is equivalent to simply looking at the histogram of the raw values of the variable of interest.

5.2.3.2 Actual Test

To conduct a one-sample t-test:

1. Analyze -> Compare Means -> One-Sample T-Test
2. Put the variable of interest in the `Test Variable(s)` box
3. Test value can be any value technically but for our application here it is set at a value of 1
4. Check off `Estimate effect sizes`
5. Click `OK`

In the output, you should see descriptive statistics (mean, SD) for hte variable of interest in the first table. The second table provides the actual significance test for the one-sample t-test. Finally, the third table provides effect sizes for the difference between the sample mean and the hypothetical mean.

5.2.4 Nonparametrics/Robust

If it is the case that the assumptions for the t-tests or the one-way ANOVA appear to be substantially violated, then there are a few alternate statistical methods that can be conducted that are more robust.

5.2.4.1 Violations of Constant Variance

The Welch's family of tests was already discussed in the case of a violation of the **constant variance** assumption, and as a matter of fact the Welch's tests are recommended as the default forms of independent-samples t-tests and one-way ANOVAs. One issue with Welch's t-test, however, is that it is not robust to violations of **normality** at smaller sample sizes.

5.2.4.2 Violations of Normality

When the assumption of **normality** is indeed substantially violated, then one option is to calculate bootstrapped confidence intervals. For independent-samples t-tests, this can be done by clicking the `Bootstrap` option in the dialogue box, inputting the number of bootstrapped samples and a particular random seed number, and clicking `Continue`. The same procedure can be done with one-way ANOVAs through the `Bootstrap` option in its own dialogue box.

Another option in the case of a independent-samples t-test is to conduct a Mann-Whitney U test. This is a non-parametric test, and this test is somewhat conceptually similar to the independent-samples t-test, in that two independent groups are being compared on some outcome variable. A misconception with the Mann-Whitney U test is that it *also* tests equality of means between two groups like an independent-samples t-test does, or that it tests equality of medians. Rather, this test more simply looks at something similar to the idea of the probability of superiority effect size - it tests whether the probability of an observation from one group being larger/smaller than an observation from a second group is different from a coin-flip chance (i.e., 50%). One issue with the Mann-Whitney U test is that it makes the assumption of *constant variance*, and therefore if it is thought that both normality and constant variance are violated, then this test would not be appropriate. Arguably, it would be appropriate only if the normality assumption was violated.

Another option in the case of a one-way ANOVA is to conduct a Kruskal-Wallis test. This test has the same misconceptions as the Mann-Whitney U test, and has the same limitation where it assumes constant variance among the groups. Again, if it seems to be the case that normality is violated *only*, then this test might be another option for comparing an outcome variable between three or more groups (while noting it does not explicitly test differences in means).

In SPSS, the Mann-Whitney U and Kruskal-Wallis tests can be conducted with the following steps:

1. Analyze -> Nonparametric Tests -> Independent Samples
2. Go to Settings -> Customize Tests -> Click the tests of interest
3. Go to Fields -> Put the outcome variable under `Test Fields` and the grouping variable under `Groups`

5.3 Sample Size Planning

`GPower` can be used to conduct power analyses for independent-samples t-tests (see the links provided in the Regression section of the document).

There are a few other online resources that can be used for doing a power analysis for t-tests and one-way ANOVAs:

- https://designingexperiments.shinyapps.io/power_ttest2group/ (an online app for independent-samples t-test)
- https://designingexperiments.shinyapps.io/BUCSS_ss_power_it/ (another online app)
- https://designingexperiments.shinyapps.io/BUCSS_ss_power_ba/ (for one-way ANOVAs)
- https://journals.sagepub.com/doi/full/10.1177/2515245920951503 (another app for ANOVAs)
- https://www.taylorfrancis.com/books/mono/10.4324/9781315171500/applied-power-analysis-behavioral-sciences-christopher-aberson

6 ANCOVA

6.1 Background

In the following examples, we will make use of the familiar `SymptomData.csv` dataset that has been used for previous statistical methods. The three variables that we will be interested in are the depression diagnosis variable (`dep_dx`), the social phobia total score variable (`sp_total`), and the relationship quality total score variable (`rq_total`).

6.2 SPSS

First, the **linearity assumption** (i.e., a linear relation between the covariate and the outcome within both groups) is assessed via plotting with scatterplots:

1. Graphs -> Chart Builder
2. Go to Scatter/Dot and drag the first plot into the preview box
3. Drag rq_total (the outcome) to the Y-axis, sp_total (the covariate) to the X-axis, and drag dep_dx (the grouping variable) to the 'Set Color?' box
4. Under Linear Fit Lines, select 'Subgroups'; click OK
5. Double-click the resulting graph, select 'Elements' at the top of SPSS, and click Fit Line at Subgroups (to get LOESS curves)

Second, the **homogeneity of slopes assumption** can be assessed with the same plot that was just created by looking at the straight lines and see if they are roughly parallel. To conduct a hypothesis test to determine whether this assumption is violated with respect to statistical signifiance:

1. Analyze -> General Linear Model -> Univariate
2. Put rq_total under `Dependent Variable`, sp_total under `Covariate(s)`, and dep_dx under `Fixed Factor(s)`
3. Click `Model` -> Click Build Terms -> Select dep_dx in the left box and click the right-pointing arrow in the middle (do the same for sp_total) -> Highlight both dep_dx and sp_total in the left box and click the right arrow -> Click Continue
4. Click OK

In the output, the ANOVA table for the model should be presented, and you can check the p-value for the interaction term (`dep_dx * sp_total`).

Now, we will run the actual ANCOVA model, evaluate some more assumptions, and look at the results.

1. Analyze -> General Linear Model -> Univariate
2. Click `Model` and click the interaction term in the right box -> Click the left arrow to remove it -> Click Continue
3. Click `EM Means` -> Drag dep_dx to the right box -> Click 'Compare main effects' -> Select the Bonferroni or Sidak adjustment (if comparing more than two groups) -> Click Continue
4. Click `Save` -> Select Unstandardized Predicted Values and Standardized Residuals -> Click Continue
5. Click `Options` -> Select Descriptive statistics, estimates of effect size, homogeneity tests, and residual plots -> Click Continue -> Click OK

Before we look at the actual results of the model, we need to evaluate the **normality assumption**. The standardized residuals that were saved in the previous sequence of steps will be analyzed via plots and hypothesis tests:

1. Analyze -> Descriptive Statistics -> Explore
2. Put the Standardized Residual variable under `Dependent List`
3. Click `Plots` -> Click 'Normality plots with tests' and 'Histogram' -> Click Continue
4. Click OK

We then need to evaluate the **constant variance assumption**. Levene's test is provided in the output from the General Linear Model dialogue box. To evaluate the assumption graphically:

1. Graphs -> Chart Builder
2. Put `PRE` variable on x-axis and `ZRE` variable on y-axis
3. Click `Groups/Point ID` and select 'Column panel variable'
4. In the new box that pops up in the chart preview (labeled 'Panel?'), drag the grouping variable into it
5. Click OK

Note that outliers can be assessed on the residuals using methods discussed in previous lectures and supplemental notes (e.g., z-score greater than |3|; robust median rule).

Finally, onto the actual results of the ANCOVA. The first output of interest is the 'Tests of Between-Subjects Effects', where hypothesis tests are included to determine whether there are significant mean differences between groups, controlling for the covariate (refer to the row with the grouping variable). The second output of interest is the Estimated Marginal Means. The Estimates table provides *estimated* group means on the outcome *holding the covariate fixed at a value* (i.e., these are adjusted group means after controlling for the covariate). The Pairwise Comparisons table provides the formal hypothesis test of group differences on the outcome, again while controlling for the covariate (i.e., are the two or more groups significantly different from each other on the outcome, controlling for the covariate?).

6.3 Sample Size Planning

For `G*Power`, see the following slides: https://med.und.edu/research/daccota/_files/pdfs/berdc_resource_pdfs/sample_size_gpower_module.pdf

The following articles may also be helpful for determining sample sizes in ANCOVA designs:

- https://www.sciencedirect.com/science/article/abs/pii/S0895435607000613
- https://link.springer.com/article/10.1007/s11336-019-09692-3
- https://www.tandfonline.com/doi/abs/10.1080/00273171.2016.1219841

7 Two-Way/Factorial ANOVA

7.1 Background

In the following example, we will use the familiar `SymptomData.csv` dataset yet again. The three variables that we will be interested in are the depression diagnosis variable (`dep_dx`), the anxiety diagnosis variable (`gad_dx`), and the relationship quality total score variable (`rq_total`). The two 'diagnosis' variables are binary variables indicating the presence or absence of the given disorder.

7.2 SPSS

We will estimate a factorical/two-way ANOVA using our data. This model may also be labeled as a 2x2 ANOVA model because each of the categorical independent variables (dep_dx and gad_dx) have *two levels* (i.e., presence or absence of diagnosis). The dependent variable in this model will be relationship quality.

First, we will check for any possible violations in the **normality** assumption:

1. Analyze -> Descriptives -> Explore
2. Place the outcome variable (rq_total) in the 'Dependent List' box; place the independent variables (dep_dx and gad_dx) in the 'Factor List' box.
3. Click `Plots` -> Select 'Histogram' and 'Normality plots with tests' -> Click `Continue`
4. Click `OK`

The output should display histograms and QQ-plots of relationship quality scores at each level of the independent variables, along with Shapiro-Wilk's tests. If it is the case that it appears that normality is substantially violated, you can perform bootstrapping as an additional step in the next series of steps.

To now actually run a factorial/two-way ANOVA:

1. Analyze -> General Linear Model -> Univariate
2. Place the outcome variable (rq_total) under 'Dependent Variable' and place the two independent variables under 'Fixed Factor(s)'
3. Click `Plots` -> put either dep_dx or gad_dx in Horizontal Axis -> put the opposite variable in Separate Lines -> Click 'Add' and Click 'Continue'
4. Click `EM Means` -> drag the three terms to the right box (dep_dx, gad_dx, and dep_dx*gad_dx) -> select 'Compare main effects' and 'Compare simple main effects' -> select a confidence interval adjustment if necessary -> Click `Continue`
5. Click `Save` -> select Studentized Residuals -> Click `Continue`
6. Click `Options` -> select 'Descriptive Statistics', 'Estimates of effect sizes', and 'Homogeneity tests' -> Click `Continue`
7. Click `OK`

The output should display multiple tables. The second table 'Descriptive Statistics' are descriptive statistics for the outcome variable stratified by group status. The third table 'Levene's Test of Equality of Error Variances' shows a significance test for the **assumption of constant variance**. In this example, Levene's test is statistically significant. To correct for this possible violation of constant variance, you can either: (a) go back into Step 6 above and additionally select 'Parameter estimates with robust standard errors' or (b) you can add an extra step where click `Bootstrap` and select the number of bootstraps, the seed, and the percentile confidence intervals. Going back to the output, the fourth table 'Tests of Between-Subjects Effects' presents the main effects of the two independent variables and the interaction between the two variables.

Post-hoc tests can be found under the 'Estimated Marginal Means' section of the output. If it is the case that the interaction is significant, I **do not** suggest interpreting the main effect post-hoc tests as they have a convoluted interpretation in the presence of a significant interaction. Rather, I would focus on the interaction post-hoc tests which tell you two different things: (a) the first table shows you how the difference in relationship quality between depressed and non-depressed individuals further depends on whether those same individuals have or don't have a simultaneous anxiety diagnosis, and (b) the second table shows you how the difference in relationship quality between anxious and non-anxious individuals further depends on whether those same individuals have or don't havea a simultaneous depression diagnosis. If it is the case that the interaction term is not significant, then you could interpret the main effects as usual (i.e., as 'adjusted' mean differences on a diagnosis controlling for the other diagnosis).

As usual, outliers can also be checked on the saved studentized residuals by using the z > |3| rule.

7.3 Reporting Example

For this first reporting example, we will assume that the interaction between anxiety and depression is *not* of interest, and we just want to interpret the main effects of anxiety and depression on relationship quality:

"A two-way between-subjects ANOVA was conducted in order to test the null hypotheses of no main effect of depression or anxiety on relationshup quality. The depression and anxiety independent variables contained two levels: having the diagnosis or not having the diagnosis. Descriptive statistics for relationship quality scores by diagnosis status are displayed in Table X. There were/were not any outliers on studentized residuals, using a z-score cut-off of |3|. With respect to the assumption of constant variance (aka homogeneity of variance), Levene's Test of Equality of Error Variances was conducted, and it was found that the null hypothesis of constant variance was rejected, F(3, 721) = 7.17, $p < .001$. Because there was evidence of the constant variance assumption being violated, robust standard errors (HC3) were used for the tests of the main effects. The overall model was statistically significant, F(3, 721) = 22.17, $p < .001$, partial eta-squared = .084. The main effect of depression diagnosis was found to be statistically significant, F(1, 721) = 37.91, $p <$ .001, partial eta-squared = .05, while the main effect of anxiety was found to not be statistically significant, F(1, 721) = .34, $p = .558$, partial eta-squared = .00. Post-hoc tests for the main effect of depression were conducted using estimated marginal means. The difference in relationship quality between those with depression and those without depression, controlling for anxiety diagnosis, was statistically significant such that those without depression had better relationship quality than those with depression (*Mdiff* = 4.39, *SE* = .71, $p < .001$, 95% CI [2.99, 5.79])."

In the above, you could also report the mean difference statistics between those with and without an anxiety diagnosis for the sake of transparency, even though the main effect of anxiety was not statistically significant in the first place. As well, you could also report the estimated marginal means for each group in addition to the mean difference only.

For this second reporting example, we will now assume that the interaction between depression and anxiety diagnosis is of interest, and therefore we just want to interpret the effects of the interaction. Although with the particular dataset that we're using the interaction is not statistically significant, we will assume it is for the sake of demonstrating how to report the results of a significant interaction.

"A two-way ANOVA was conducted in order to test the null hypothesis of no interaction between depression diagnosis and anxiety diagnosis on relationshup quality scores. Descriptive statistics for relationship quality scores by group are displayed in Table X. There were/were not any outliers on studentized residuals, using a z-score cut-off of |3|. With respect to the assumption of constant variance (aka homogeneity of variance) Levene's Test was conducted, and it was found that the null hypothesis of constant variance was rejected, F(3, 721) = 7.17, $p < .001$. Because there was evidence of the constant variance assumption being violated, robust standard errors (HC3) were used for the test of the interaction. The overall model was statistically significant, F(3, 721) = 22.17, $p < .001$, partial eta-squared = .084. The interaction between depression and anxiety was not statistically significant, F(1, 721) = .078, $p = .781$, partial eta-squared = .00. Post-hoc tests for the interaction of depression with anxiety were conducted using estimated marginal means. First, it was found that the difference in relationship quality between depressed and non-depressed individuals was less pronounced if those same individuals had anxiety (*Mdiff* = -4.19, *SE* = 1.04, $p < .001$) compared to if those same individuals did not have anxiety (*Mdiff* = -4.59, *SE* = .98, $p < .001$). Second, flipping the interaction around, it was also found that the difference in relationship quality between anxious and non-anxious individuals was more pronounced if those individuals did not have depression (*Mdiff* = -.62, *SE* = .90, $p = .492$) compared to if those individuals did have depression (*Mdiff* = -.22, *SE* = 1.11, $p = .843$). A plot of this interaction can be found in Figure X."

7.4 Sample Size Planning

GPower is able to conduct power analyses for factorial designs, see the following two sets of slides:

- https://core.ecu.edu/wuenschk/docs30/GPower3-ANOVA-Factorial.pdf
- https://med.und.edu/research/daccota/_files/pdfs/berdc_resource_pdfs/sample_size_gpower_module.pdf

The documentation on the GPower website may also be useful.

8 Repeated Measures

8.1 Background

In the following examples, we will make use of a simulated dataset called `DepressionTreatment.csv` that has the following structure:

- Participants were randomized to receive either a novel treatment (n = 100) or were assigned to a waitlist control (n = 100). The `Group` variable in the dataset represents which group a participant was assigned to. In other words, there is a between-subjects variable of group.
- Participants were assessed on their depression symptoms (ranging from 0 to 27) at three different time points. They were assessed at baseline, at post-treatment, and at a follow-up time point. The scores for each person at each of these time points can be found in the `Baseline`, `PostTreatment`, and `FollowUp` variables, respectively.

As mentioned during lecture, many typical examples of the methods in this document involve data that are structured like the above. That is, participants may provide data over *multiple time points*. The other common design that is relevant is one where participants go through a *sequence of experimental conditions*. That is, participants provide data for each experimental condition, meaning that they have data points for *each* level of the categorical independent variable 'condition'.

8.2 SPSS

8.2.1 Paired T-Test

When conducting a paired t-test, it is helpful to know in advance which two time points (or conditions) you would like to compare. In the following example, we will compare whether there is a significant difference/change between scores taken at Baseline compared to scores taken at Follow-Up. It's important to note that we are collapsing across treatment group (`Group` variable), in the sense that we are **not** particularly interested in whether the change from baseline to follow-up is dependent on which treatment group someone is in. We just want to know, regardless of treatment group, whether the average score changed from baseline to follow-up. If you want to know whether the change does depend on treatment group, then you have to run a Mixed ANOVA (see the Mixed ANOVA section of this document).

To conduct a paired t-test in SPSS:

1. Analyze -> Compare Means -> `Paired-Samples T Test`
2. Place the two variables corresponding to Baseline scores and Follow-Up scores in the right box, one by one
3. If you would like to correct for possible violations of **constant variance**, you can click `Bootstrap`, select `Perform bootstrapping`, and set a seed for reproducibility; Click `Continue`
4. Click `OK`

There should be several tables that are in the output. The *first* table 'Paired Samples Statistics' displays the means and SDs of baseline scores and follow-up scores. The *second* table presents the correlation between baseline and follow-up scores, which is usually not of interest for this test but interesting nonetheless. The* *third* table 'Paired Samples Test' provides the actual significance test for the difference between baseline and follow-up scores. The column 'Means' has the unstandardized mean difference between the two time points, and the column 'Significance' has the p-value associated with the significance test (focus on the two-sided p-value). Finally, the *fourth* table 'Paired Samples Effect Sizes' displays the Cohen's d effect size in the 'Point Estimate' column.

8.2.1.1 Reporting Example Here is an example of how to report the results we got from the paired t-test we just ran:

"A paired-samples t-test was conducted in order to test the null hypothesis of no change in depression scores between Baseline and Follow-Up. Bootstrapped confidence intervals (based on 1000 bootstrapped samples) using the percentile method were also calculated to account for possible violations of assumptions. Results suggested that Follow-Up scores (M = 21.39, SD = 3.43) were significantly lower than Baseline scores (M = 24.09, SD = .91), p < .001, with a mean difference of 2.70, bootstrapped 95% CI [1.99, 3.45], and Cohen's D = .77, 95% CI [.54, .99]."

Note that the means and standard deviations were taken from the 'Paired Samples Statistics' table in the output, the p-value and bootstrapped CIs were taken from the 'Bootstrap for Paired Samples Test' table, and the Cohen's D was taken from the 'Paired Samples Effect Sizes' table.

8.2.2 Repeated-Measures ANOVA

In the following example, we will test whether or not there is some effect of time on depression scores. That is, do depression scores significantly change over time and, if so, between *which* time points?

To conduct a repeated-measures ANOVA:

1. Analyze -> General Linear Model -> Repeated Measures
2. Under 'Within-Subject Factor Name', enter a variable name for the categorical independent variable you are interested in. In this case we are interested in the effect of time, so you can name the variable 'time' (if you were interested in condition effects then you could name the variable 'condition')
3. Beside 'Number of Levels' enter the number of time points (or conditions) you are looking at. In this case, we are looking at three time points, so the number 3 is put in
4. Click `Add`, then click `Define`
5. Drag the variables corresponding to time points 1, 2, and 3 (in the correct order; e.g., 'Baseline' first, 'PostTreatment' second, etc.) to the Within-Subjects Variables box
6. Click `Plots` -> drag 'time' to Horizontal Axis -> Click `Add` -> Click `Continue`
7. Click `EM Means` -> draft 'time' to the 'Display means for' box -> Click `Compare main effects` -> Select a confidence interval adjustment if you want -> Click `Continue`
8. Click `Save` -> select 'Studentized' Residuals -> Click `Continue`
9. Click `Options` -> select 'Descriptive statistics' and 'Estimates of effect size' -> Click `Continue`
10. Click `OK`

There should be several tables in the output. Looking at the *second* table 'Descriptive Statistics' is where the average scores at each time point can be found. The *fourth* table 'Mauchly's Test of Sphericity' is where the assumption of constant variance is tested (i.e., it is called sphericity in the context of repeated-measures). Recall from lecture that if this test is significant and the Greenhouse-Geiser value is less than .70, then it is likely that constant variance is violated. In our case, it seems like it is. The *fifth* table 'Tests of Within-Subjects Effects' provides the global test for the effect of time on depression scores containing the p-value and the effect size of partial eta squared. Because constant variance/sphericity was violated, we should be looking at the 'Greenhouse-Geiser' row.

If we scroll down to the 'Estimated Marginal Means' section of the output, this is where the actual post-hoc tests are done to determine *which* time points are significantly different from each other. The 'Pairwise Comparisons' table provides the significance tests between each pair of time points. If you selected LSD under the confidence interval adjustment in Step 7 above, then every time point seems to be significantly different from each other.

A plot of the scores over time can be found at the bottom of the output.

To assess for **outliers**, we can take a look at whether there are participants that have a z-score of greater than |3| on the newly saved Studentized Residuals variables (labeled as `SRE` in the dataset).

To assess for **normality**, we can go through the Analyze -> Descriptive Statistics -> Explore menu, drag our variables to the right, and select 'Normality plots with tests' and 'Histogram' under the `Plots` button.

8.2.2.1 Reporting Example

Here is an example of how to report the results of the repeated-measures ANOVA that we just ran:

"A repeated-measures ANOVA was conducted in order to test the null hypothesis of no change in depression scores between any of the three time points. That is, time was considered to be the within-subjects factor containing three levels, and depression scores were considered to be the dependent variable. There were/were not any outliers on studentized residuals, using a z-score cut-off of |3|. With respect to the assumption of sphericity, Mauchly's Test was statistically significant ($\chi(2) = 98.33$, $p < .001$) and the Greenhouse-Geiser epsilon value was .612 (less than the recommended .70 cut-off). Therefore, the Greenhouse-Geiser correction was applied to the main effect of time. The main effect of time was found to be statistically significant, F(1.224, 121.224) = 60.881, $p < .001$, partial eta-squared = .38. Post-hoc tests of the main effect of time were conducted using estimated marginal means with no correction for multiple comparisons. Scores at baseline (M = 24.09, SE = .09) were found to be significantly larger than scores at post-treatment (M = 23.68, SE = .19, p = .041) and scores at follow-up (M = 21.39, SE = .34, $p < .001$). Scores at follow-up were also significantly lower than scores at post-treatment, $p < .001$."

Note that you can optionally report the unstandardized mean *differences* (along with the confidence intervals around these differences) rather than the means for each time point on their own.

8.2.3 Mixed ANOVA

In the following example, we will estimate a Mixed ANOVA model involving the main effects of time and treatment group, and the interaction between time and treatment group. The main effect of 'time' attempts to answer the question of "Regardless of which treatment group someone is in, do depression scores change over the three time points?" (notice that this is very much like the question we asked in the repeated-measures ANOVA above). The main effect of 'group' attempts to answer the question of "Averaged across all of the time points, do those in the treatment group tend to have more/less depression than those in the control group?" The interaction between 'time' and 'group' attempts to answer the following two questions: (a) "Is the change in depression between the various time points more pronounced depending on whether you are in the treatment group or control group?", and (b) "Does the cross-sectional difference in depression between those in the treatment group and control group depend on which time point you look at?" Typically, question (a) lines up with research questions involved in longitudinal studies, whereas question (b) lines up more with research questions from experimental condition studies.

To conduct a mixed ANOVA:

1. Analyze -> General Linear Model -> Repeated Measures
2. Under 'Within-Subject Factor Name', enter a variable name for the within-person categorical independent variable you are interested in. In this case we are interested in the effect of time within participants, so you can name the variable 'time' (if you were interested in condition effects then you could name the variable 'condition')
3. Beside 'Number of Levels' enter the number of time points (or conditions) you are looking at. In this case, we are looking at three time points, so the number 3 is put in
4. Click `Add`; Click `Define`
5. Drag the variables corresponding to time points 1, 2, and 3 (in the correct order; e.g., 'Baseline' first, 'PostTreatment' second, etc.) to the Within-Subjects Variables box
6. Drag the between-subjects variable of 'group' to the Between-Subjects Factor box (i.e., the variable where participants can only be a part of one level, in this case *either* treatment or control)
7. Click `Plots` -> drag 'time' to Horizontal Axis and 'Group' to Separate Lines -> Click `Add` (and optionally select 'Include Error bars') and click `Continue`
8. Click `EM Means` -> drag 'time', 'Group', and 'Group*time' to the 'Display Means for:' box -> select both 'Compare main effects' and 'Compare simple main effects' -> select a confidence interval adjustment -> Click `Continue`
9. Click `Save` -> select Studentized Residuals -> Click `Continue`
10. Click `Options` -> select 'Descriptive statistics' and 'Estimates of effect size' -> Click `Continue`
11. Click `OK`

There should be several tables in the output. The *third* table 'Descriptive Statistics' displays the means of depression scores at each time point and by each group. The *fifth* table 'Mauchly's Test of Sphericity' is where the assumption of constant variance is tested (i.e., it is called sphericity in the context of mixed models). If this test is statistically significant and the Greenhouse-Geiser value is less than .70, then there is evidence that the constant variance assumption is violated. In our case, it does not seem to be the case. The *sixth* table 'Tests of Within-Subjects Effects' displays the main effect for 'time' and the interaction between 'time' and 'Group', along with associated effect sizes. The table labeled 'Tests of Between-Subjects Effects' displays the main effect for 'Group' along with an associated effect size.

If we scroll down to the 'Estimated Marginal Means' section of the output, this is where we can find the post-hoc comparisons for each pair of time points and each pair of Groups (which in our case there is only one pair). Importantly for a mixed ANOVA are the 'Group * time' subsections, which basically probe the interaction effect. For example, the change from time 1 to time 2 in the Control group is -1.153 (going up) while the change from time 1 to time 2 is 1.984 in the Treatment group.

A plot of the scores over time can be found at the bottom of the output.

To assess for **outliers**, we can take a look at whether there are participants that have a z-score of greater than |3| on the newly saved Studentized Residuals variables (labeled as `SRE` in the dataset).

To assess for **normality**, we can go through the Analyze -> Descriptive Statistics -> Explore menu. In our case, because we are testing a within-subjects variable (time) with a between-subjects variable (group) interaction, we want to look at each time point stratified by group status. We would first drag the time variables to the 'Dependent List' box, and then drag the Group variable to the 'Factor List' box. Finally, we would select 'Normality plots with tests' and 'Histogram' under the `Plots` button.

8.2.3.1 Reporting Example A 2x3 mixed ANOVA was conducted in order to test the null hypothesis of no interaction between group (i.e., treatment vs control) and time (i.e., baseline, post-treatment, follow-up) on depression scores. That is, group was considered to be the between-subjects factor containing two levels, time was considered to be the within-subjects factor containing three levels, and depression was the outcome. Descriptive statistics for depression scores by time and group are displayed in Table X. There were/were not any outliers on studentized residuals, using a z-score cut-off of |3|. With respect to the assumption of sphericity, Maichly's Test was not statistically significant ($\chi(2) = 2.471$, $p = .291$) and the Greenhouse-Geiser epsilon value was .975 (above the recommended .70 cut-off). Therefore, sphericity was assumed for the interaction between time and group. The interaction between time and group was found to be statistically significant, F(2, 196) = 305.864, $p < .001$, partial eta-squared = .76. Post-hoc tests of the main effect of time were conducted using estimated marginal means with no correction for multiple comparisons. A plot of the interaction effect can be found in Figure X. For the interaction, first, results suggested that the difference in depression scores between the treatment group and the control group depended on time. That is, the mean difference between the treatment and control group at Baseline was not statistically significant (*Mdiff* = -.11, *SE* = .18, $p = .542$), while the mean difference between the treatment and control group was statistically significant at Post-Treatment (*Mdiff* = -3.25, *SE* = .18, $p < .001$) and at Follow-Up (*Mdiff* = -6.56, *SE* = .19, $p < .001$). Second, the results also suggested that the *change* in depression scores *between time points* depended on whether someone was in the treatment group or in the control group. The change from Baseline to Post-Treatment in the control group was actually positive (*Mdiff* = 1.15, *SE* = .18, $p < .001$) while the change was negative in the treatment group (*Mdiff* = -1.98, *SE* = .18, $p < .001$). The change from Post-Treatment in the control group (*Mdiff* = -.63, *SE* = .18, $p < .001$) was also less pronounced than the change in the treatment group (*Mdiff* = -3.93, *SE* = .18, $p < .001$). And lastly the change from Baseline to Follow-Up was actually positive in the control group (*Mdiff* = .53, *SE* = .20, $p = .009$) while the change was negative in the treatment group (*Mdiff* = -5.92, *SE* = .20, $p < .001$).

Note that you can also include the confidence intervals around each mean difference if you have space in your document. Here, I omitted it for the sake of length.

8.3 Sample Size Planning

GPower is able to conduct power analyses for mixed ANOVAs, please see the following slides for how to conduct this: https://med.und.edu/research/daccota/_files/pdfs/berdc_resource_pdfs/sample_size_gpower_module.pdf. The documentation on the website for GPower may also be helpful.

More online resources are the following: - https://designingexperiments.shinyapps.io/BUCSS_ss_power_wa/ (online app for mixed ANOVA) - https://www.taylorfrancis.com/books/mono/10.4324/9781315171500/applied-power-analysis-behavioral-sciences-christopher-aberson